(i) A ray passing through the first principal focus will, after refraction, emerge parallel to the axis [see ray 1 in Figs. 4.8(a) and (b)]

(ii) A ray parallel to the axis will, after refraction, either pass through or appear to come from (depending on the sign of f_2) the second principal focus [see ray 2 in Figs. 4.8(a) and (b)]

(iii) A ray passing through the center of the lens P will pass through undeviated* [see ray 3 in Figs. 4.8(a) and (b)]

4.6 THE NEWTON FORMULA

LO 3

Let x_1 be the distance of the object from the first principal focus F_1 (x_1 will be positive if the object point is on the right of F_1 and conversely) and let x_2 be the distance of the image from the second principal focus F_2 as shown in Figs. 4.8(a) and (b). Considering similar triangles in Fig. 4.8(a), we have

$$\frac{-y'}{y} = \frac{-f_1}{-x_1} \qquad \textbf{(4.17)}$$

and

$$\frac{-y'}{y} = \frac{x_2}{f_2} \qquad \textbf{(4.18)}$$

where the vertical distances are positive if measured above the line and negative if measured below the line (see Sec. 4.2.1). Equations (4.17) and (4.18) give

$$f_1 f_2 = x_1 x_2 \qquad \textbf{(4.19)}$$

which is known as the *Newtonian lens formula*. It may be noted that for a diverging lens [see Fig. 4.8(b)], Eqs. (4.17) and (4.18) would be

$$\frac{y'}{y} = \frac{f_1}{-x_1} = \frac{x_2}{-f_2}$$

which are identical to Eqs. (4.17) and (4.18).

When the thin lens has the same medium on the two sides, then using Eq. (4.13) we have

$$x_1 x_2 = -f^2 \qquad \textbf{(4.20)}$$

showing that x_1 and x_2 must be of opposite signs. Thus, if the object lies on the left of the first principal focus, then the image will lie on the right of the second principal focus, and vice versa.

4.7 LATERAL MAGNIFICATION

The lateral magnification m is the ratio of the height of the image to that of the object. Considering either Fig. 4.8(a) or (b) we readily get

$$m = \frac{y'}{y} = \frac{v}{u} = \frac{f_2 + x_2}{f_1 + x_1} = -\frac{f_1}{x_1} = -\frac{x_2}{f_2} \qquad \textbf{(4.21)}$$

where we have made use of Eqs. (4.17) and (4.18). Obviously, if m is positive, the image is erect [as in Fig. 4.8(b) and conversely if m is negative, the image is inverted as in Fig. 4.8(a)].

The magnification can also be calculated as the product of the individual magnifications produced by each of the

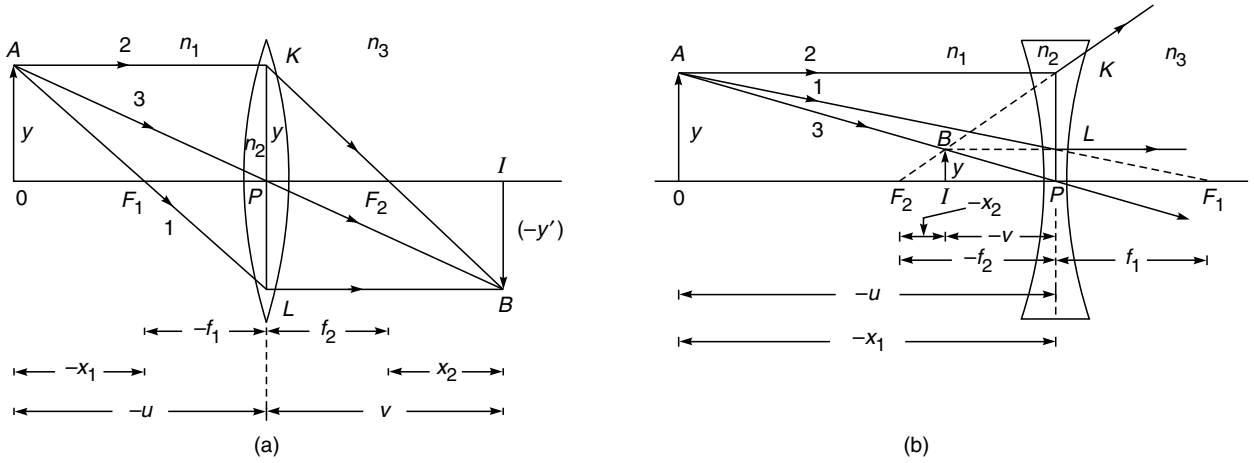


Fig. 4.8 (a) Paraxial imaging by a converging lens; x_1, f_1 and u are negative quantities and x_2, f_2, and v are positive quantities. (b) Paraxial imaging by a diverging lens; here x_1, f_2, u and v are negative quantities and x_2 and f_1 are positive quantities.

* This follows from the fact that, for a thin lens, when $u = 0$, v is also equal to zero [see Eqs. (4.10) and (4.14)].

refracting surfaces; referring to Fig. 4.9, the magnification produced by a single refracting surface is given by

$$m = \frac{y'}{y}$$

and considering triangles AOC and ICB, we get

$$\frac{-y'}{y} = \frac{v - R}{-u + R} = \frac{\frac{v}{R} - 1}{-\frac{u}{R} + 1} \qquad \textbf{(4.22)}$$

Now, Eq. (4.5) gives us

$$\frac{n_2}{n_1} - \frac{v}{u} = \frac{n_2 - n_1}{n_1}\frac{v}{R}$$

and

$$\frac{u}{v} - \frac{n_1}{n_2} = \frac{n_2 - n_1}{n_2}\frac{u}{R}$$

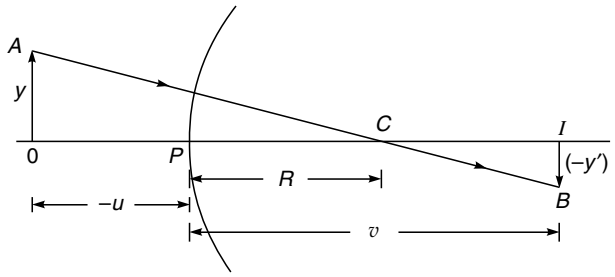


Fig. 4.9 Imaging of an object of height y by a spherical refracting surface.

Substituting for v/R and u/R in Eq. (4.22), we get

$$m = \frac{y'}{y} = \frac{n_1 v}{n_2 u} \qquad \textbf{(4.23)}$$

Thus, if m_1 and m_2 represent the magnifications produced by the two refracting surfaces in Fig. 4.8, then

$$m = \frac{n_1}{n_2}\frac{v'}{u}$$

and

$$m_2 = \frac{n_2}{n_1}\frac{v}{v'}$$

where v' represents the distance of the image formed by the first refracting surface. Thus,

$$m = m_1 m_2 = \frac{v}{u} \qquad \textbf{(4.24)}$$

consistent with Eq. (4.21).

Example 4.3 Consider a system of two thin lenses as shown in Fig. 4.10. The convex lens has a focal length of +20 cm and the concave lens has a focal length of –10 cm. The two lenses are separated by 8 cm. For an object of height 1 cm (at a distance of 40 cm from the convex lens), calculate the position and size of the image. (The same problem will be solved again in Chapter 5 by using the matrix method.)

Solution: Let us first calculate the position and size of the image formed by the first lens:

$$u = -40 \text{ cm}, f = +20 \text{ cm}$$

Therefore, using Eq. (4.11), we get

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{40} + \frac{1}{20} = +\frac{1}{40}$$

Thus, $v = +40$ cm and $m_1 = -1$; the image is of the same size but inverted. This image acts as a virtual object for the concave lens with $u = +32$ cm and $f = -10$ cm. Thus,

$$\frac{1}{v} = \frac{1}{32} - \frac{1}{10} = -\frac{22}{320}$$

giving

$$v \simeq -14.5 \text{ cm}$$

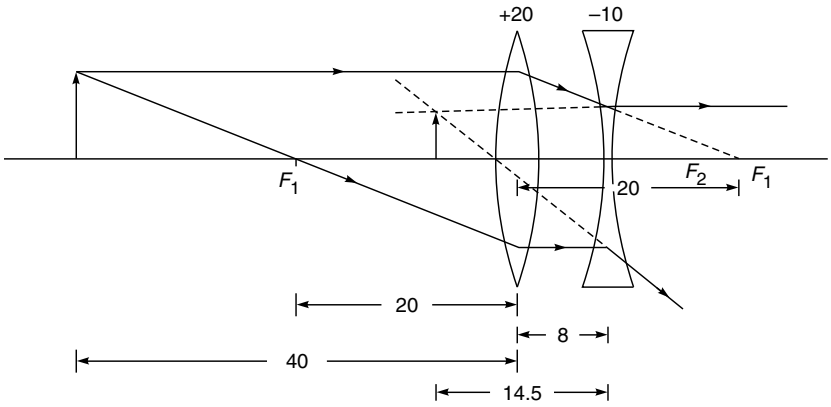


Fig. 4.10 Paraxial imaging by an optical system consisting of a converging lens of focal length 20 cm and a diverging lens of focal length -10 cm separated by 8 cm. All distances in the figure are in centimeters.